

Research Analysis Report

Research Topic: Agentic AI

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'Agentic AI' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey', 'year': 2025, 'summary': 'Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.'}

• {'title': 'AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges', 'year': 2025, 'summary': 'This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for task-specific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. Through a chronological evaluation of architectural evolution, operational mechanisms, interaction styles, and autonomy levels, we present a comparative analysis across both AI agents and agentic AI paradigms. Application domains enabled by AI Agents such as customer support, scheduling, and data summarization are then contrasted with Agentic AI deployments in research automation, robotic coordination, and

medical decision support. We further examine unique challenges in each paradigm including hallucination, brittleness, emergent behavior, and coordination failure, and propose targeted solutions such as ReAct loops, retrieval-augmented generation (RAG), automation coordination layers, and causal modeling. This work aims'}

• {'title': 'Small Language Models are the Future of Agentic AI', 'year': 2025, 'summary': 'Large language models (LLMs) are often praised for exhibiting near-human performance on a wide range of tasks and valued for their ability to hold a general conversation. The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation. Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI. Our argumentation is grounded in the current level of capabilities exhibited by SLMs, the common architectures of agentic systems, and the economy of LM deployment. We further argue that in situations where general-purpose conversational abilities are essential, heterogeneous agentic systems (i.e., agents invoking multiple different models) are the natural choice. We discuss the potential barriers for the adoption of SLMs in agentic systems and outline a general LLM-to-SLM agent conversion algorithm. Our position, formulated as a value statement, highlights the significance of the operational and economic impact even a partial shift from LLMs to SLMs is to have on the AI agent industry. We aim to stimulate the'}

• {'title': 'The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges', 'year': 2025, 'summary': 'Agentic AI systems are a recently emerged and important approach that goes beyond traditional AI, generative AI, and autonomous systems by focusing on autonomy, adaptability, and goal-driven reasoning. This study provides a clear review of agentic AI systems by bringing together their definitions, frameworks, and architectures, and by comparing them with related areas like generative AI, autonomic computing, and multi-agent systems. To do this, we reviewed 143 primary studies on current LLM-based and non-LLM-driven agentic systems and examined how they support planning, memory, reflection, and goal pursuit. Furthermore, we classified architectural models, input–output mechanisms, and applications based on their task domains where agentic AI is applied, supported using tabular summaries that highlight real-world case studies. Evaluation metrics were classified as qualitative and quantitative measures, along with available testing methods of agentic AI systems to check the system's performance and reliability. This study also highlights the main challenges and limitations of agentic AI, covering technical, architectural, coordination, ethical, and security issues. We organized the conceptual foundations, available tools, architectures, and evaluation metrics in this research, which defines a structured foundation for understanding and advancing agentic AI. These findings aim to help researchers and developers build better, clearer, and more adaptable systems'}

• {'title': 'Agentic AI for Scientific Discovery: A Survey of Progress, Challenges, and Future Directions', 'year': 2025, 'summary': 'The integration of Agentic AI into scientific discovery marks a new frontier in research automation. These AI systems, capable of reasoning, planning, and autonomous decision-making, are transforming how scientists perform literature review, generate hypotheses, conduct experiments, and analyze results. This survey provides a comprehensive overview of Agentic AI for scientific discovery, categorizing existing systems and tools, and highlighting recent progress across fields such as chemistry, biology, and materials science. We discuss key evaluation metrics, implementation frameworks, and commonly used datasets to offer a detailed understanding of the current state of the field. Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration.'}

Research Trends

- ability
- adaptability
- agent
- agentic

- analysis
- application
- artificial
- automation
- autonomous
- autonomy
- between
- beyond
- capable
- challenge
- complex
- conceptual
- critically
- decision-making
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- scientific
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- traditional
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- unlike
- valued
- wide

Research Gaps

- Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration
- Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI
- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts

Future Scope

- Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration
- Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI

- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

- {'title': 'Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey', 'research_area': 'Agentic AI', 'research_problem': 'Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention', 'methodology': ['Survey', 'Framework', 'System'], 'keywords': ['agentic', 'artificial', 'intelligence', 'refer', 'autonomous', 'system', 'designed', 'pursue', 'complex', 'goals', 'minimal', 'human', 'intervention', 'unlike', 'traditional'], 'year': 2025, 'citation_count': 639, 'quality_score': 100, 'quality_classification': 'Excellent'}
- {'title': 'AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges', 'research_area': 'Agentic AI', 'research_problem': 'This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities', 'methodology': ['Model', 'System', 'Llm', 'Rag'], 'keywords': ['review', 'critically', 'distinguish', 'between', 'agent', 'agentic', 'offering', 'structured', 'conceptual', 'taxonomy', 'application', 'mapping', 'analysis', 'opportunity', 'challenge'], 'year': 2025, 'citation_count': 472, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'Small Language Models are the Future of Agentic AI', 'research_area': 'Agentic AI', 'research_problem': 'Large language models (LLMs) are often praised for exhibiting near-human performance on a wide range of tasks and valued for their ability to hold a general conversation', 'methodology': ['Architecture', 'Model', 'System', 'Algorithm', 'Llm'], 'keywords': ['large', 'language', 'model', 'llms', 'often', 'praised', 'exhibiting', 'near-human', 'performance', 'wide', 'range', 'tasks', 'valued', 'ability', 'hold'], 'year': 2025, 'citation_count': 332, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges', 'research_area': 'Agentic AI', 'research_problem': 'Agentic AI systems are a recently emerged and important approach that goes beyond traditional AI, generative AI, and autonomous systems by focusing on autonomy, adaptability, and goal-driven reasoning', 'methodology': ['Framework', 'Architecture', 'Model', 'System', 'Approach', 'Llm'], 'keywords': ['agentic', 'system', 'recently', 'emerged', 'important', 'goes', 'beyond', 'traditional', 'generative', 'autonomous', 'focusing', 'autonomy', 'adaptability', 'goal-driven', 'reasoning'], 'year': 2025, 'citation_count': 142, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'Agentic AI for Scientific Discovery: A Survey of Progress, Challenges, and Future Directions', 'research_area': 'Agentic AI', 'research_problem': 'The integration of Agentic AI into scientific discovery marks a new frontier in research automation', 'methodology': ['Survey', 'Framework', 'System'], 'keywords': ['integration', 'agentic', 'scientific', 'discovery', 'marks', 'frontier', 'automation', 'system', 'capable', 'reasoning', 'planning', 'autonomous', 'decision-making', 'transforming', 'scientist'], 'year': 2025, 'citation_count': 102, 'quality_score': 90, 'quality_classification': 'Excellent'}

Highest Cited Paper

title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

authors: ['D. Acharya', 'Karthigeyan Kuppan', 'Divya Bhaskaracharya']

year: 2025

citation_count: 639

summary: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.

research_problem: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention

methodology: ['Survey', 'Framework', 'System']

key_contributions: ['This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI', 'This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively']

future_work: ['We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts']

keywords: ['agentic', 'artificial', 'intelligence', 'refer', 'autonomous', 'system', 'designed', 'pursue', 'complex', 'goals', 'minimal', 'human', 'intervention', 'unlike', 'traditional']

research_area: Agentic AI

paper_score: 100

paper_quality: Excellent

Latest Paper

title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

authors: ['D. Acharya', 'Karthigeyan Kuppan', 'Divya Bhaskaracharya']

year: 2025

citation_count: 639

summary: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a

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research_problem: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention

methodology: ['Survey', 'Framework', 'System']

key_contributions: ['This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI', 'This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively']

future_work: ['We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts']

keywords: ['agentic', 'artificial', 'intelligence', 'refer', 'autonomous', 'system', 'designed', 'pursue', 'complex', 'goals', 'minimal', 'human', 'intervention', 'unlike', 'traditional']

research_area: Agentic AI

paper_score: 100

paper_quality: Excellent

Common Methodologies

- Algorithm
- Approach
- Architecture
- Framework
- Llm
- Model
- Rag
- Survey
- System

Research Areas

- Agentic AI

Common Keywords

- ability
- adaptability
- agent
- agentic
- analysis
- application
- artificial

- automation
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Methodology Frequency

System: 5

Framework: 3

Model: 3

Llm: 3

Survey: 2

Architecture: 2

Rag: 1

Algorithm: 1

Approach: 1

Most Common Methodology: System

Quality Distribution

Excellent: 5

Publication Years

2025: 5

Citation Statistics

highest: 639

lowest: 102

average: 337.4

total: 1687

Comparison Summary

total_papers: 5

most_common_methodology: System

research_areas: ['Agentic AI']

quality_distribution: {'Excellent': 5}

citation_statistics: {'highest': 639, 'lowest': 102, 'average': 337.4, 'total': 1687}

highest_cited_title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

latest_paper_title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Research Gap Analysis

Total Papers: 5

Research Areas

- Agentic AI

Common Keywords

- ability
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Future Work

- Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration
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Research Area Distribution

Agentic AI: 5

Keyword Frequency

agentic: 4
autonomous: 3
system: 3
traditional: 2
reasoning: 2
artificial: 1
intelligence: 1
refer: 1
designed: 1
pursue: 1
complex: 1
goals: 1

minimal: 1
human: 1
intervention: 1
unlike: 1
review: 1
critically: 1
distinguish: 1
between: 1
agent: 1
offering: 1
structured: 1
conceptual: 1
taxonomy: 1
application: 1
mapping: 1
analysis: 1
opportunity: 1
challenge: 1
large: 1
language: 1
model: 1
llms: 1
often: 1
praised: 1
exhibiting: 1
near-human: 1
performance: 1
wide: 1
range: 1
tasks: 1
valued: 1
ability: 1
hold: 1
recently: 1
emerged: 1
important: 1
goes: 1
beyond: 1

generative: 1
focusing: 1
autonomy: 1
adaptability: 1
goal-driven: 1
integration: 1
scientific: 1
discovery: 1
marks: 1
frontier: 1
automation: 1
capable: 1
planning: 1
decision-making: 1
transforming: 1
scientist: 1

Research Trends

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- autonomous
- system
- traditional
- reasoning
- artificial
- intelligence
- refer
- designed
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- complex
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- minimal
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Gap Categories

datasets: []

models: ['Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI']

evaluation: []

applications: []

general: ['Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration', 'We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts']

Emerging Topics

- agentic
- autonomous
- system
- traditional
- reasoning

Recommendations

- Explore autonomous multi-agent systems.
- Improve reasoning capabilities using LLM planning.

Summary

dominant_area: Agentic AI

top_trend: agentic

future_work_items: 3

Experiment Plan

Research Areas

- Agentic AI

Recommended Datasets

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- OpenAI Evals

Baseline Models

None

Evaluation Metrics

- Accuracy

- Precision
- Recall
- F1-Score
- ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

- Train-Test Split
- K-Fold Cross Validation
- Hyperparameter Tuning
- Statistical Significance Testing

Experimental Workflow

- Collect Dataset
- Preprocess Data
- Train Baseline Models
- Train Proposed Model
- Evaluate Performance
- Compare Results
- Analyze Errors
- Draw Conclusions

Report Summary

Dominant Research Area: Agentic AI

Top Research Trend: agentic

Highest Cited Paper: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Latest Paper: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Recommended Datasets

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets

- OpenAI Evals

Baseline Models

None

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.